☐ CTL Provisional DECISION SHEET (PDSH)
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Standard(s) (incl. year)	Subclause(s)	Tracking No.
IEC 62893-4-1:2020	Cl. 4.3.1 Table B.1 Table B.2	DSH 2239
Category		
CABL		
Subject	**Keywords**	**Developed by**
Requirements for the cable with split power cores.	Split power cores	ETF 6

Question

It is stated in Note b of Table B.1 in IEC 62893-4-1:2020, that "Power cores could be split in two smaller cores".

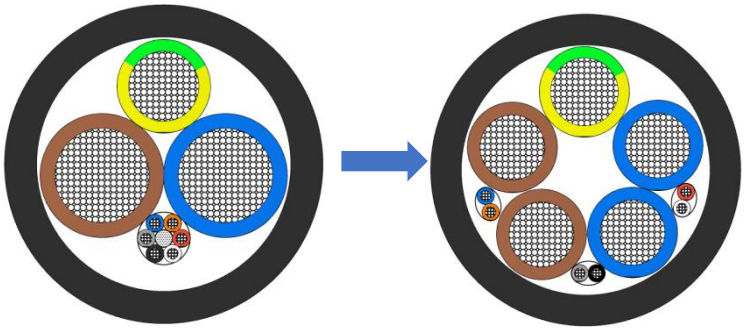


Fig. 1 Typical structural images of cable before and after splitting power cores

As it's getting more popular to split the power cores into smaller cross-sections due to the roundness of cable as illustrated in Fig. 1, the following questions need to be addressed and clarified to better guide the design and test of those cables.

1) Are the split power cores required to have the same nominal cross-sectional area?

2) What are the specific requirements for those split power cores?

Decision

1) According to Cl. 4.3.1 and note b of Table B.1/B.2 in IEC 62893-4-1:2020, power core is only allowed to be split into two smaller cores with the same nominal cross-sectional area, which shall be the one listed in Table B.1/B.2 in IEC 62893-4-1:2020. Some examples of split power cores are provided in the following table.

No.	Size of cables	
	Cable before splitting power cores	Cable after splitting power cores
1	2X95	4X50
2	2X70	4X35
3	2X50	4X25
4	2X35	4X16

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2) Some specific requirements of split power cores are as follows. a) Conductor resistance, Insulation resistance, Diameter of wires in conductor, Thickness of insulation The same with that of power core with the same nominal cross-sectional area in Table B.1/B.2. b) Bending test for cables with conductor cross section larger than 4 mm² The cross-section of power cores refers to the one of original power cores, instead of the split power cores, when determining the mechanical load for bending test.
Explanatory notes
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